

■ 2.4.4 Functional Composition

When you manipulate functions as objects in their own right, an operation you often need to perform is functional composition.

<code>Compose</code> $[f_1, f_2, \dots, f_k, x]$	compose expressions to give $f_1[f_2[\dots[f_k[x]]\dots]]$
<code>Nest</code> $[f, x, n]$	apply f composed n times to x
<code>HeadCompose</code> $[f_1, f_2, \dots, f_k, x]$	compose expressions to give $f_1[f_2]\dots[f_k][x]$
<code>Operate</code> $[p, f[x]]$	give $p[f][x]$
<code>Operate</code> $[p, f[x], n]$	apply p at level n in f

Functional composition operations.

This forms the composition of a , b and c , and applies the result to x .

```
In[1]:= Compose[a, b, c, x]
Out[1]= a[b[c[x]]]
```

You can use `Compose` with `Outer` to do this.

```
In[2]:= Outer[Compose, {a, b}, {x, y, z}]
Out[2]= {{a[x], a[y], a[z]}, {b[x], b[y], b[z]}}
```

`Nest` $[f, x, n]$ composes the function f with itself n times, and applies the result to x .

```
In[3]:= Nest[f, x, 4]
Out[3]= f[f[f[f[x]]]]
```

`Operate` performs an operation on the head of an expression.

```
In[4]:= Operate[p, a[b][x]]
Out[4]= p[a[b]][x]
```

`HeadCompose` constructs this slightly peculiar expression. Remember that function application groups to the left, so the expression should be read $((a[b])[c])[x]$.

```
In[5]:= HeadCompose[a, b, c, x]
Out[5]= a[b][c][x]
```